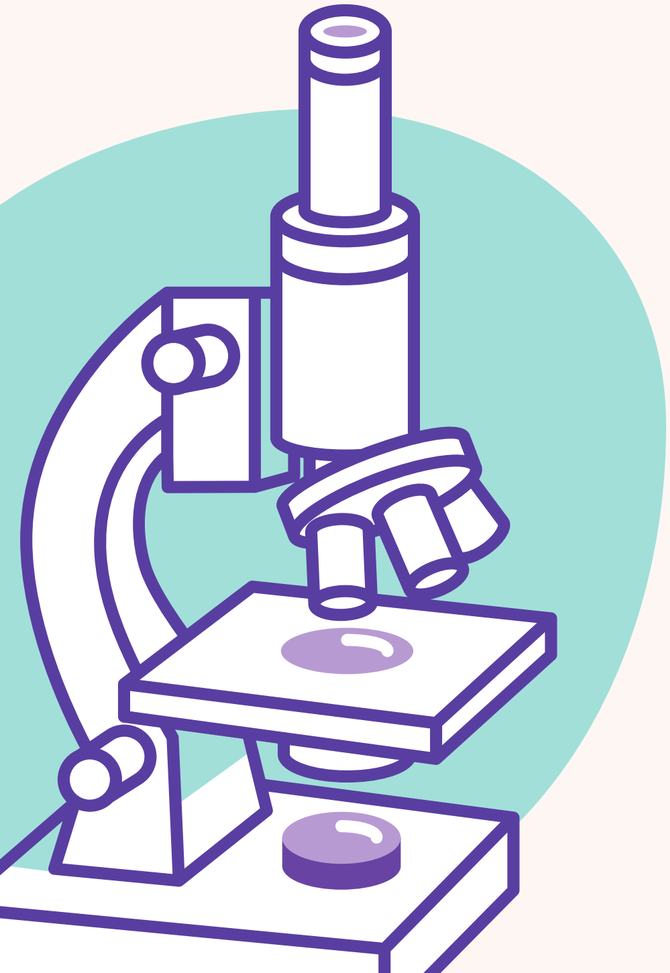
Group 14 Elements





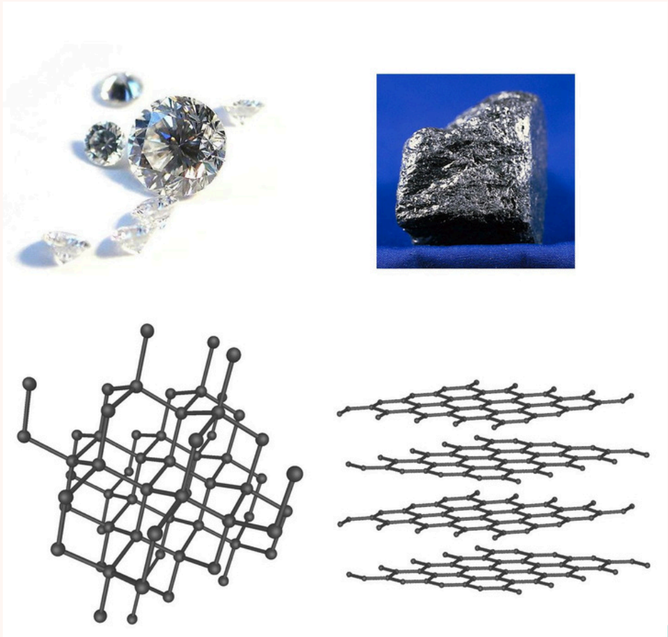
Hybridization of Carbon

Carbon monoxide is poisonous CO forms a stable complex with haemoglobin called carboxy haemoglobin.This prevents haemoglobin from carrying oxygen into the cells of the body and it ultimately results death.



Allotropy

The phenomenon of existence of an element in different physical forms having same chemical properties is called allotropy.



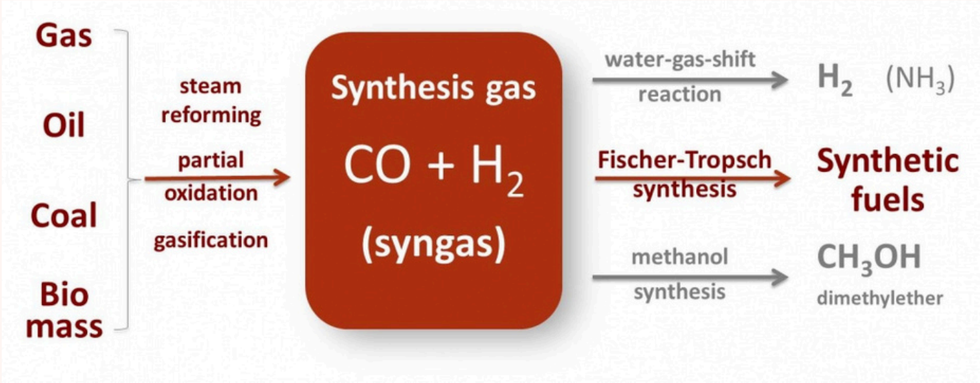
Graphite

In Graphite, each carbon atom is in sp2 hybridised state.
Each carbon atom contains one electron in pure 'p' orbital.
Due to the presence of these free electrons, graphite acts as a good conductor of electricity.

Graphite function as a lubricant.
It has two dimensional layer structure and these layers can easily slide one over the other,as they are held by weak vanderwaal forces.

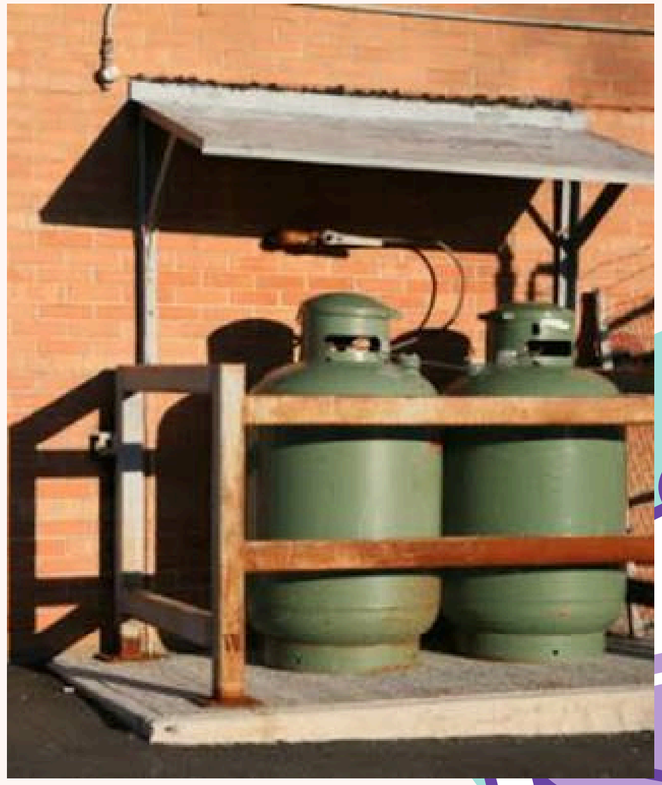
Synthesis Gas

The mixture ofcarbon monoxide and hydrogen
is known as water gas or synthesis gas or syngas.
It is used for the synthesis of methanol
and number of hydrocarbons.



Producer Gas

The mixture of carbon monoxide and nitrogen is known as producer Gas.
It is prepared by passing air over hot cake.



Silicones

They are organo silicon polymers in which silicon is strongly linked with oxygen and carbon .

